Regularizing and Coding Summable Ideals

Candidate full solutions to Questions 6.1 and 6.2 of arXiv:2011.10638

Autonomous proof candidate for human review

11 August 2026

Status. Candidate full affirmative solutions, likely valid. Every summable ideal generated by an element of $\mathscr{Z}$ is isomorphic to a single ideal generated by an element of $\mathscr{Y}$, answering Question 6.1 more strongly than requested. Moreover, the set $\mathscr{A}$ is analytic in $\mathscr{Z}$, answering Question 6.2.

1 Question and answer

For a positive sequence $x = (x_n)$ with $\sum_n x_n = \infty$, put

$$\mathcal{I}_x = \left\{ A \subseteq \mathbb{N} : \sum_{n \in A} x_n < \infty \right\}.$$

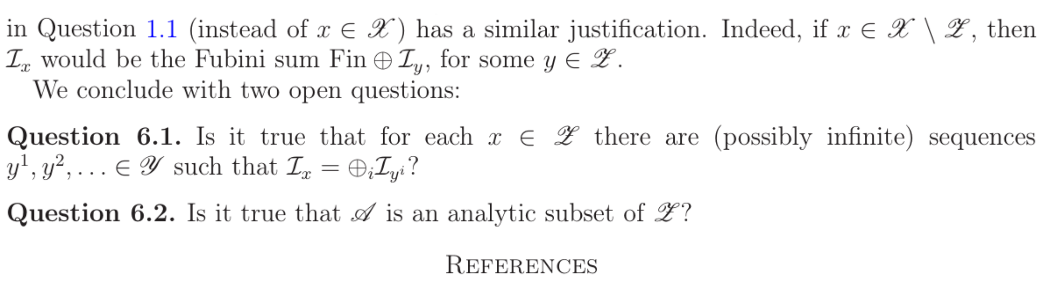
Following the notation of [1], let

$$\mathscr{Z} = \{x \in c_0 : x_n > 0 \text{ for every } n, \ \sum_n x_n = \infty\}, \qquad \mathscr{Y} = \left\{ y \in \mathscr{X} : \liminf_n \frac{y_{n+1}}{y_n} > 0 \right\},$$

and

$$\mathscr{A} = \{x \in \mathscr{Z} : \mathcal{I}_x \neq \mathcal{I}_y \text{ for every } y \in \mathscr{Y}\}.$$

Question 6.1 asks whether every $\mathcal{I}_x$ is, up to the ideal-isomorphism convention used immediately before the question, a finite or countable direct sum of ideals generated by elements of $\mathscr{Y}$. Question 6.2 asks whether $\mathscr{A}$ is analytic in $\mathscr{Z}$.



in Question 1.1 (instead of $x \in \mathscr{X}$) has a similar justification. Indeed, if $x \in \mathscr{X} \setminus \mathscr{Z}$, then $\mathcal{I}_x$ would be the Fubini sum $\mathrm{Fin} \oplus \mathcal{I}_y$, for some $y \in \mathscr{Z}$.

We conclude with two open questions:

Question 6.1. Is it true that for each $x \in \mathscr{Z}$ there are (possibly infinite) sequences $y^1, y^2, \ldots \in \mathscr{Y}$ such that $\mathcal{I}_x = \oplus_i \mathcal{I}_{y^i}$?

Question 6.2. Is it true that $\mathscr{A}$ is an analytic subset of $\mathscr{Z}$?

References

Figure 1: The two concluding questions in the source, PDF page 6. This packet resolves both.

For $A \subseteq \mathbb{N}$, write

$$DA = \{n - 1 : n \in A, \ n \geq 2\}.$$

The central observation is the following exact witness formula.

Theorem 1. *For every $x \in \mathscr{X}$,*

$$x \in \mathscr{A} \quad \Longleftrightarrow \quad \exists A \subseteq \mathbb{N} \quad A \in \mathcal{I}_x \text{ and } DA \notin \mathcal{I}_x. \tag{1}$$

Consequently $\mathscr{A}$ is an analytic subset of $\mathscr{X}$.

The right side of (1) is visibly an existential real witness. The substance is that no other obstruction to regularizing the weights exists.

2 A comparison lemma for summable ideals

We first isolate the elementary diagonal-selection argument that supplies a uniform estimate modulo a summable exceptional set.

Lemma 1. *Let $u = (u_n)$ and $v = (v_n)$ be positive null sequences. Then*

$$\mathcal{I}_u \subseteq \mathcal{I}_v$$

if and only if there is a constant $C > 0$ for which

$$\sum_{\{n : v_n > C u_n\}} v_n < \infty. \tag{2}$$

Proof. If (2) holds and $A \in \mathcal{I}_u$, split A into the exceptional set and its complement to obtain

$$\sum_{n \in A} v_n \leq \sum_{\{n : v_n > C u_n\}} v_n + C \sum_{n \in A} u_n < \infty.$$

Conversely, suppose (2) fails for every C. Inductively choose disjoint finite sets F_k, supported after all previously chosen indices, such that

$$F_k \subseteq \{n : v_n > 2^k u_n\}, \qquad 1 \leq \sum_{n \in F_k} v_n \leq 2.$$

This is possible because the relevant v-sum is infinite even after a finite initial segment is removed, and $v_n \to 0$ lets the last selected term be at most 1. For $A = \bigcup_k F_k$,

$$\sum_{n \in A} u_n < \sum_{k \geq 1} 2^{-k} \sum_{n \in F_k} v_n \leq 2 \sum_{k \geq 1} 2^{-k} < \infty,$$

whereas $\sum_{n \in A} v_n \geq \sum_k 1 = \infty$. Thus $A \in \mathcal{I}_u \setminus \mathcal{I}_v$, proving the contrapositive. □

3 Backward-shift invariance is exactly regularizability

Lemma 2. *For $x \in \mathscr{X}$, the following are equivalent:*

1. *there exists $y \in \mathscr{Y}$ with $\mathcal{I}_x = \mathcal{I}_y$;*
2. *$A \in \mathcal{I}_x$ implies $DA \in \mathcal{I}_x$ for every $A \subseteq \mathbb{N}$.*

Proof. Assume 1, and choose $\eta > 0$ and N such that $y_n \geq \eta y_{n-1}$ for $n \geq N$. If $A \in \mathcal{I}_y$, then

$$\sum_{n\in A,\ n\geq N} y_{n-1} \leq \eta^{-1} \sum_{n\in A,\ n\geq N} y_n < \infty.$$

The omitted initial part is finite, so $DA \in \mathcal{I}_y$. Equality of the two ideals gives 2.

Now assume 2. Define $v_1 = x_1$ and $v_n = x_{n-1}$ for $n \geq 2$. Then 2 says precisely that $\mathcal{I}_x \subseteq \mathcal{I}_v$ (the first coordinate is harmless). Lemma 1 supplies $C > 0$ such that, for

$$S = \{n \geq 2 : x_{n-1} > Cx_n\}, \qquad \sum_{n\in S} x_{n-1} < \infty.$$

Choose $0 < \kappa < \min\{1, C^{-1}\}$ and recursively set

$$y_1 = x_1, \qquad y_n = \max\{x_n, \kappa y_{n-1}\} \quad (n \geq 2). \tag{3}$$

Thus $y_n/y_{n-1} \geq \kappa$ for every $n \geq 2$.

It remains to check that the regularization changes the weights by a summable amount. Put $d_n = y_n - x_n \geq 0$. If $n \notin S$, then $x_{n-1} \leq Cx_n$, so (3) gives

$$d_n = \max\{0, \kappa(x_{n-1} + d_{n-1}) - x_n\} \leq \kappa d_{n-1},$$

because $\kappa C < 1$. If $n \in S$, the same formula gives $d_n \leq \kappa d_{n-1} + \kappa x_{n-1}$. Hence, in all cases,

$$d_n \leq \kappa d_{n-1} + \kappa x_{n-1}\mathbf{1}_S(n). \tag{4}$$

Summing (4) up to an arbitrary index M and using $d_1 = 0$ yields

$$(1-\kappa)\sum_{n=1}^{M} d_n \leq \kappa \sum_{n\in S} x_{n-1}.$$

Therefore $\sum_n d_n < \infty$. Since $y = x + d$ with d summable,

$$\sum_{n\in A} x_n < \infty \quad \Longleftrightarrow \quad \sum_{n\in A} y_n < \infty$$

for every $A \subseteq \mathbb{N}$. Also $y \geq x$ makes $\sum_n y_n = \infty$, and $y_n \to 0$ follows from $x_n \to 0$ and $d_n \to 0$. Thus $y \in \mathscr{Y}$ and $\mathcal{I}_x = \mathcal{I}_y$. □

Lemma 2 immediately proves the equivalence in (1): membership in $\mathscr{A}$ is the negation of item 1 of Lemma 2, hence the negation of its universal shift condition in item 2.

4 Every summable ideal has one regular summand

We now obtain a stronger affirmative answer to Question 6.1. As in the paragraph preceding that question in [1], equality with a direct sum is understood after an isomorphic identification of the underlying countable sets.

Theorem 2. *For every $x \in \mathscr{X}$, there are a permutation σ of $\mathbb{N}$ and a sequence $y \in \mathscr{Y}$ such that, with $u_n = x_{\sigma(n)}$,*

$$\mathcal{I}_u = \mathcal{I}_y.$$

Consequently $\mathcal{I}_x$ is isomorphic to the one-summand direct sum $\mathcal{I}_y$.

Proof. Because $x_n \to 0$ and all terms are positive, its terms can be listed in nonincreasing order: greedily select a largest unused term. A largest term exists at every stage, and every original index is eventually selected since only finitely many terms are at least as large as its positive value. This gives a permutation σ for which

$$u_1 \geq u_2 \geq \cdots > 0, \qquad u_n \to 0, \qquad \sum_n u_n = \infty.$$

Let

$$S = \{n \geq 2 : u_{n-1} > 2u_n\}.$$

If $n_1 < n_2 < \cdots$ enumerate S, monotonicity gives

$$u_{n_{j+1}-1} \leq u_{n_j} < \tfrac{1}{2} u_{n_j-1}.$$

Thus the predecessor weights at the large drops form a geometrically decreasing sequence, and

$$\sum_{n \in S} u_{n-1} < \infty.$$

Apply Lemma 1 to u_n and $v_1 = u_1$, $v_n = u_{n-1}$ for $n \geq 2$, with $C = 2$. It follows that $\mathcal{I}_u \subseteq \mathcal{I}_v$, or equivalently that $\mathcal{I}_u$ is invariant under D. Lemma 2 now produces $y \in \mathscr{Y}$ with $\mathcal{I}_u = \mathcal{I}_y$.

Finally, $A \mapsto \sigma[A]$ is an isomorphism from $\mathcal{I}_u$ onto $\mathcal{I}_x$. Hence one regular summand suffices. □

5 Descriptive-set verification

Identify $\mathcal{P}(\mathbb{N})$ with Cantor space $2^{\mathbb{N}}$. For $m \geq 1$, define continuous functions on $2^{\mathbb{N}} \times \mathscr{X}$ by

$$s_m(A, x) = \sum_{\substack{n \leq m \\ n \in A}} x_n, \qquad t_m(A, x) = \sum_{\substack{2 \leq n \leq m \\ n \in A}} x_{n-1}.$$

Then the witness set

$$W = \left\{ (A, x) : \sum_{n \in A} x_n < \infty \text{ and } \sum_{n \in DA} x_n = \infty \right\}$$

is Borel, since it equals

$$\left(\bigcup_{K \geq 1} \bigcap_{m \geq 1} \{s_m \leq K\} \right) \cap \left(\bigcap_{K \geq 1} \bigcup_{m \geq 1} \{t_m > K\} \right).$$

By Theorem 1,

$$\mathscr{A} = \{x \in \mathscr{X} : \exists A \in 2^{\mathbb{N}}\ (A, x) \in W\}.$$

This is the projection of a Borel subset of the product of two Polish spaces, and is therefore analytic. This completes the proof.

6 Checks and scope

The comparison lemma is sharp enough to handle sparse, arbitrarily deep drops in the weights: only the total mass of their predecessors must be summable. The geometric envelope (3) then propagates each repair with ratio κ, while (4) prevents the total repair from diverging.

Theorem 2 uses the isomorphism interpretation explicitly adopted in the source paragraph introducing Question 6.1. If the displayed equality there were instead read as literal equality on the original fixed coordinates, one summand would be exactly the false assertion already disproved by the source; a direct sum on a different countable base necessarily requires an identification.

A bounded search of the run indexes, the exact question text, arXiv records on summable ideals, and papers indexed as citing [1] located no later resolution of either concluding question. Priority is not asserted; specialist review is required before dissemination.

References

[1] M. Balcerzak and P. Leonetti, *Convergent subseries of divergent series*, Results Math. **77** (2022), article 84; arXiv:2011.10638.